

Benchmark results — Persistences, BLEND, AR-OLS, and ELM cost-function variants (Palaiseau)

The runs use $N_{\text{hidden}} = 500$ and $N_{\text{draws}} = 100$, $\approx 35\,000$ samples (2 years, Aug 2020 – Aug 2022).

The ELM and AR-OLS inputs are the standardized window of the $LB = 48$ lags, augmented by 4 cyclical time features (sin/cos encoding of the hour of the day and of the day of the year) for the target time step.

Predictions are clipped to $[0, +\infty)$. RMSE is in W.

The maximum value observed over this window is $\approx 16,484$ W: the RMSE values below should be read relative to this peak (e.g. an RMSE of $1,648$ W is $\approx 10\%$ of it).

Each ELM variant selects its hyperparameter(s) from a small grid by **expanding-window temporal cross-validation** (5 folds): the value(s) minimizing the mean validation RMSE over the folds are kept, then β is refit on the whole training set. The λ column reports the selected value (or tuple, in the order stated in each variant’s script). **Bold** marks the best NICE ^{Σ} among the cyclic ELM variants for each table (ties bolded together).

All data vs. daytime only. Each horizon is reported twice: *All data* includes all 30 min test steps (nights, where $P \approx 0$, dominate in count) and *Daytime only*, which restricts the evaluation to steps where the solar elevation at the site (48.71°N , 2.22°E) exceeds 0° .

Rolling ELM. The **ELM** (**{Box-Cox,Ridge}** roll. **{1m,6m,1y}**) rows report full-window rolling variants (Box-Cox and Ridge cost functions) at three training-window sizes (1 month, 6 months, 1 year): the model is trained on *every* 30 min supervised pair of the sliding window $[t - W, t]$, with β re-solved once per week (the hidden layer re-drawn per refit and λ chosen per horizon by temporal CV). **Their test set is not the one used by the static models:** the rolling scheme needs one year of history before the first prediction, so all windows are evaluated on the same later, shorter window (aligned on the 1-year history requirement). Their NICE values are therefore *indicative* and the block carries its own **bold** (best NICE ^{Σ} across all rolling variants and windows), separate from the static models above.

FH = 0.5 h

Table 1: All data – $FH = 0.5$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1169.0	0.490	~ 0	0.239	0.910	0.999	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.793	1.966	1.938	1.899
BLEND _{corr}	0	–	–	–	1850.1	0.775	+0.001	0.352	0.775	1.469	1.583	1.550	1.534
BLEND _{optim}	48	–	–	–	1079.7	0.452	−0.014	0.216	0.923	0.901	0.924	0.931	0.918
AR-OLS	53	–	–	–	1008.4	0.422	+0.001	0.200	0.933	0.835	0.863	0.880	0.859
ELM (OLS)	27000	500	100	–	979.6	0.410	−0.001	0.190	0.937	0.792	0.838	0.861	0.830
ELM (Ridge)	27000	500	100	25	974.0	0.408	−0.003	0.189	0.938	0.791	0.833	0.853	0.826
ELM (Rob. Risk)	27000	500	100	5	974.0	0.408	−0.003	0.189	0.938	0.791	0.833	0.853	0.826
ELM (Tikhonov)	27000	500	100	25	974.1	0.408	−0.002	0.189	0.938	0.789	0.833	0.854	0.826
ELM (Box-Cox)	27000	500	100	(10,0.75)	978.6	0.410	−0.020	0.186	0.937	0.777	0.837	0.863	0.826
ELM (MAE)	27000	500	100	(25,500)	977.3	0.409	~ 0	0.184	0.937	0.767	0.836	0.869	0.824
ELM (Log-MSE)	27000	500	100	(25,100)	1226.0	0.514	−0.043	0.234	0.901	0.979	1.049	1.069	1.032
ELM (Huber)	27000	500	100	25	977.9	0.410	~ 0	0.183	0.937	0.766	0.837	0.870	0.824
ELM (L3)	27000	500	100	(10,500)	1115.6	0.467	−0.002	0.239	0.918	0.999	0.954	0.925	0.960
ELM (Corr)	27000	500	100	(25,0.5)	989.1	0.414	−0.001	0.182	0.936	0.761	0.846	0.889	0.832
ELM (Elastic Net)	27000	500	100	(10,25)	974.0	0.408	−0.003	0.189	0.938	0.791	0.833	0.853	0.826
ELM (M-Est.)	27000	500	100	25	973.1	0.408	−0.003	0.181	0.938	0.757	0.832	0.867	0.819
ELM (Lp)	27000	500	100	(25,1.25)	971.6	0.407	−0.003	0.186	0.938	0.777	0.831	0.857	0.822
ELM (GLM)	27000	500	100	(25,0.1)	1070.8	0.448	+0.024	0.218	0.925	0.910	0.916	0.924	0.917
ELM (Box-Cox roll. 1m)	27000	–	–	(50,0.25)	1133.3	0.474	−0.050	0.222	0.916	0.926	0.967	0.979	0.957
ELM (Box-Cox roll. 6m)	27000	–	–	(25,0.75)	1001.2	0.418	−0.008	0.198	0.934	0.829	0.854	0.879	0.854
ELM (Box-Cox roll. 1y)	27000	–	–	(25,0.75)	988.3	0.413	−0.016	0.195	0.936	0.813	0.843	0.867	0.841
ELM (Ridge roll. 1m)	27000	–	–	25	1018.4	0.426	~ 0	0.201	0.932	0.841	0.869	0.882	0.864
ELM (Ridge roll. 6m)	27000	–	–	10	993.4	0.415	+0.003	0.192	0.935	0.802	0.848	0.872	0.840
ELM (Ridge roll. 1y)	27000	–	–	25	980.2	0.410	−0.005	0.190	0.937	0.795	0.836	0.859	0.830
ELM _{nocyc} (Ridge)	25000	500	100	25	992.6	0.416	+0.016	0.208	0.935	0.869	0.849	0.866	0.861
ELM _{nocyc} (Rob.R.)	400	8	2	3.16228	2392.8	0.798	+0.077	0.576	0.650	1.749	1.297	1.152	1.400
NeuralProphet	–	–	–	–	1005.9	0.420	−0.022	0.227	0.934	0.947	0.859	0.874	0.893
NeuralProphet _{no seas.}	–	–	–	–	1314.1	0.549	−0.008	0.347	0.887	1.450	1.122	1.059	1.210

Table 2: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 0.5$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1650.0	0.347	~ 0	0.239	0.857	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.795	1.966	1.938	1.900
BLEND _{corr}	0	–	–	–	2611.3	0.549	+0.001	0.352	0.642	1.470	1.583	1.550	1.534
BLEND _{optim}	48	–	–	–	1524.0	0.320	−0.014	0.216	0.878	0.902	0.924	0.931	0.919
AR-OLS	53	–	–	–	1423.3	0.299	+0.001	0.200	0.894	0.836	0.863	0.880	0.859
ELM (OLS)	27000	500	100	–	1382.7	0.291	−0.001	0.189	0.900	0.792	0.838	0.861	0.830
ELM (Ridge)	27000	500	100	25	1374.8	0.289	−0.003	0.189	0.901	0.792	0.833	0.853	0.826
ELM (Rob. Risk)	27000	500	100	5	1374.8	0.289	−0.003	0.189	0.901	0.792	0.833	0.853	0.826
ELM (Tikhonov)	27000	500	100	25	1374.9	0.289	−0.002	0.189	0.901	0.790	0.833	0.854	0.826
ELM (Box-Cox)	27000	500	100	(10,0.75)	1381.3	0.290	−0.019	0.186	0.900	0.778	0.837	0.863	0.826
ELM (MAE)	27000	500	100	(25,500)	1379.4	0.290	~ 0	0.183	0.900	0.767	0.836	0.869	0.824
ELM (Log-MSE)	27000	500	100	(25,100)	1730.5	0.364	−0.043	0.234	0.843	0.980	1.049	1.069	1.033
ELM (Huber)	27000	500	100	25	1380.3	0.290	~ 0	0.183	0.900	0.766	0.837	0.870	0.824
ELM (L3)	27000	500	100	(10,500)	1574.7	0.331	−0.002	0.239	0.870	1.000	0.954	0.925	0.960
ELM (Corr)	27000	500	100	(25,0.5)	1396.0	0.294	~ 0	0.182	0.898	0.762	0.846	0.889	0.832
ELM (Elastic Net)	27000	500	100	(10,25)	1374.8	0.289	−0.003	0.189	0.901	0.792	0.833	0.853	0.826
ELM (M-Est.)	27000	500	100	25	1373.5	0.289	−0.003	0.181	0.901	0.758	0.832	0.867	0.819
ELM (Lp)	27000	500	100	(25,1.25)	1371.4	0.288	−0.003	0.186	0.901	0.777	0.831	0.857	0.822
ELM (GLM)	27000	500	100	(25,0.1)	1511.4	0.318	+0.025	0.218	0.880	0.911	0.916	0.924	0.917
ELM (Box-Cox roll. 1m)	27000	–	–	(50,0.25)	1598.8	0.336	−0.051	0.221	0.866	0.922	0.967	0.979	0.956
ELM (Box-Cox roll. 6m)	27000	–	–	(25,0.75)	1410.0	0.296	−0.015	0.191	0.896	0.800	0.853	0.879	0.844
ELM (Box-Cox roll. 1y)	27000	–	–	(25,0.75)	1392.4	0.292	−0.023	0.188	0.898	0.785	0.842	0.867	0.831
ELM (Ridge roll. 1m)	27000	–	–	25	1436.8	0.302	~ 0	0.201	0.892	0.841	0.869	0.882	0.864
ELM (Ridge roll. 6m)	27000	–	–	10	1401.5	0.294	+0.003	0.192	0.897	0.802	0.848	0.872	0.840
ELM (Ridge roll. 1y)	27000	–	–	25	1382.9	0.290	−0.005	0.190	0.900	0.795	0.836	0.859	0.830
ELM _{nocyc} (Ridge)	25000	500	100	25	1390.8	0.292	−0.002	0.190	0.899	0.797	0.843	0.866	0.835
ELM _{nocyc} (Rob.R.)	400	8	2	3.16228	2937.4	0.585	−0.012	0.487	0.500	1.479	1.230	1.125	1.278
NeuralProphet	–	–	–	–	1404.4	0.295	−0.006	0.196	0.897	0.822	0.850	0.873	0.848
NeuralProphet _{no seas.}	–	–	–	–	1772.5	0.372	+0.005	0.264	0.835	1.104	1.072	1.049	1.075

FH = 1 h

Table 3: All data – $FH = 1$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1698.6	0.711	−0.003	0.376	0.810	0.993	1.000	1.000	0.998
Persistence P ^o	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.132	1.352	1.429	1.305
BLEND _{corr}	0	–	–	–	1885.6	0.790	~ 0	0.364	0.766	0.960	1.110	1.163	1.078
BLEND _{optim}	48	–	–	–	1379.6	0.578	−0.031	0.282	0.875	0.745	0.812	0.848	0.802
AR-OLS	53	–	–	–	1270.1	0.532	+0.003	0.267	0.894	0.704	0.747	0.784	0.745
ELM (OLS)	27000	500	100	–	1190.3	0.499	−0.005	0.241	0.907	0.635	0.700	0.749	0.695
ELM (Ridge)	27000	500	100	25	1187.1	0.497	−0.006	0.241	0.907	0.636	0.699	0.745	0.693
ELM (Rob. Risk)	27000	500	100	5	1187.1	0.497	−0.006	0.241	0.907	0.636	0.699	0.745	0.693
ELM (Tikhonov)	27000	500	100	25	1187.7	0.497	−0.005	0.240	0.907	0.634	0.699	0.747	0.693
ELM (Box-Cox)	27000	500	100	(25,0.75)	1198.2	0.502	−0.029	0.238	0.906	0.627	0.705	0.756	0.696
ELM (MAE)	27000	500	100	(25,10)	1188.1	0.498	−0.005	0.237	0.907	0.625	0.699	0.750	0.692
ELM (Log-MSE)	27000	500	100	(25,100)	1408.5	0.590	−0.080	0.275	0.870	0.727	0.829	0.882	0.812
ELM (Huber)	27000	500	100	25	1190.2	0.498	−0.004	0.233	0.907	0.616	0.700	0.757	0.691
ELM (L3)	27000	500	100	(10,500)	1341.7	0.562	−0.005	0.300	0.882	0.793	0.789	0.794	0.792
ELM (Corr)	27000	500	100	(25,0.5)	1192.6	0.500	−0.002	0.234	0.907	0.617	0.702	0.766	0.695
ELM (Elastic Net)	27000	500	100	(10,25)	1187.1	0.497	−0.006	0.241	0.907	0.636	0.699	0.745	0.693
ELM (M-Est.)	27000	500	100	10	1195.6	0.501	−0.004	0.233	0.906	0.614	0.704	0.762	0.693
ELM (Lp)	27000	500	100	(25,1.25)	1187.1	0.497	−0.005	0.236	0.907	0.623	0.699	0.751	0.691
ELM (GLM)	27000	500	100	(25,0.1)	1280.2	0.536	+0.017	0.264	0.892	0.696	0.753	0.801	0.750
ELM (Box-Cox roll. 1m)	27000	–	–	(100,0.25)	1363.2	0.570	−0.079	0.269	0.878	0.709	0.800	0.857	0.789
ELM (Box-Cox roll. 6m)	27000	–	–	(25,0.75)	1218.5	0.509	−0.010	0.251	0.903	0.662	0.715	0.769	0.715
ELM (Box-Cox roll. 1y)	27000	–	–	(25,0.75)	1199.7	0.501	−0.024	0.247	0.906	0.651	0.704	0.754	0.703
ELM (Ridge roll. 1m)	27000	–	–	50	1249.5	0.522	−0.001	0.258	0.898	0.685	0.734	0.772	0.731
ELM (Ridge roll. 6m)	27000	–	–	25	1216.1	0.508	+0.001	0.247	0.903	0.656	0.714	0.763	0.711
ELM (Ridge roll. 1y)	27000	–	–	25	1191.5	0.498	−0.006	0.242	0.907	0.643	0.700	0.746	0.696
ELM _{no cyc} (Ridge)	25000	500	100	25	1218.1	0.510	+0.016	0.266	0.903	0.702	0.717	0.762	0.727
NeuralProphet	–	–	–	–	1262.6	0.527	−0.036	0.309	0.896	0.815	0.741	0.771	0.776
NeuralProphet _{no seas.}	–	–	–	–	1478.9	0.618	−0.002	0.408	0.857	1.076	0.868	0.849	0.931

Table 4: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 1$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	2397.6	0.504	−0.003	0.376	0.698	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.140	1.353	1.429	1.308
BLEND _{corr}	0	–	–	–	2661.5	0.560	~ 0	0.364	0.628	0.967	1.110	1.163	1.080
BLEND _{optim}	48	–	–	–	1947.2	0.409	−0.031	0.282	0.801	0.750	0.812	0.848	0.804
AR-OLS	53	–	–	–	1792.6	0.377	+0.004	0.267	0.831	0.709	0.748	0.784	0.747
ELM (OLS)	27000	500	100	–	1680.1	0.353	−0.005	0.240	0.852	0.639	0.701	0.749	0.696
ELM (Ridge)	27000	500	100	25	1675.6	0.352	−0.006	0.241	0.853	0.640	0.699	0.745	0.695
ELM (Rob. Risk)	27000	500	100	5	1675.6	0.352	−0.006	0.241	0.853	0.640	0.699	0.745	0.695
ELM (Tikhonov)	27000	500	100	25	1676.4	0.352	−0.005	0.240	0.853	0.639	0.699	0.747	0.695
ELM (Box-Cox)	27000	500	100	(25,0.75)	1691.2	0.356	−0.029	0.238	0.850	0.631	0.705	0.756	0.697
ELM (MAE)	27000	500	100	(25,10)	1677.0	0.353	−0.005	0.237	0.852	0.630	0.699	0.750	0.693
ELM (Log-MSE)	27000	500	100	(25,100)	1988.1	0.418	−0.080	0.275	0.793	0.731	0.829	0.882	0.814
ELM (Huber)	27000	500	100	25	1679.9	0.353	−0.004	0.233	0.852	0.620	0.701	0.757	0.692
ELM (L3)	27000	500	100	(10,500)	1893.7	0.398	−0.005	0.300	0.812	0.798	0.790	0.794	0.794
ELM (Corr)	27000	500	100	(25,0.5)	1683.4	0.354	−0.002	0.234	0.851	0.621	0.702	0.766	0.696
ELM (Elastic Net)	27000	500	100	(10,25)	1675.6	0.352	−0.006	0.241	0.853	0.640	0.699	0.745	0.695
ELM (M-Est.)	27000	500	100	10	1687.6	0.355	−0.003	0.233	0.851	0.618	0.704	0.762	0.695
ELM (Lp)	27000	500	100	(25,1.25)	1675.6	0.352	−0.005	0.236	0.853	0.628	0.699	0.751	0.692
ELM (GLM)	27000	500	100	(25,0.1)	1806.9	0.380	+0.017	0.264	0.829	0.701	0.754	0.801	0.752
ELM (Box-Cox roll. 1m)	27000	–	–	(100,0.25)	1923.0	0.404	−0.081	0.267	0.806	0.710	0.801	0.858	0.789
ELM (Box-Cox roll. 6m)	27000	–	–	(25,0.75)	1715.2	0.360	−0.020	0.242	0.846	0.642	0.714	0.768	0.708
ELM (Box-Cox roll. 1y)	27000	–	–	(25,0.75)	1690.0	0.355	−0.033	0.238	0.850	0.632	0.704	0.753	0.696
ELM (Ridge roll. 1m)	27000	–	–	50	1762.9	0.370	~ 0	0.258	0.837	0.685	0.734	0.772	0.731
ELM (Ridge roll. 6m)	27000	–	–	25	1715.6	0.360	+0.002	0.247	0.846	0.656	0.714	0.763	0.711
ELM (Ridge roll. 1y)	27000	–	–	25	1681.1	0.353	−0.006	0.242	0.852	0.643	0.700	0.746	0.696
ELM _{no cyc} (Ridge)	25000	500	100	25	1707.6	0.359	−0.006	0.244	0.847	0.649	0.712	0.762	0.708
NeuralProphet	–	–	–	–	1750.9	0.367	−0.008	0.260	0.839	0.692	0.729	0.769	0.730
NeuralProphet _{no seas.}	–	–	–	–	1955.5	0.410	+0.012	0.298	0.800	0.791	0.814	0.834	0.813

FH = 3 h

Table 5: All data – $FH = 3$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	3485.2	1.460	−0.082	0.808	0.202	0.908	0.970	0.985	0.954
Persistence P ^o	0	–	–	–	2298.5	0.963	~ 0	0.429	0.653	0.483	0.640	0.727	0.616
BLEND _{corr}	0	–	–	–	2101.2	0.880	−0.029	0.446	0.710	0.501	0.585	0.644	0.577
BLEND _{optim}	48	–	–	–	1914.1	0.802	−0.073	0.382	0.759	0.429	0.533	0.597	0.520
AR-OLS	53	–	–	–	1785.9	0.748	−0.001	0.395	0.790	0.444	0.497	0.539	0.493
ELM (OLS)	27000	500	100	–	1653.3	0.692	−0.011	0.349	0.820	0.393	0.460	0.506	0.453
ELM (Ridge)	27000	500	100	25	1651.2	0.692	−0.013	0.350	0.821	0.393	0.459	0.505	0.453
ELM (Rob. Risk)	27000	500	100	5	1651.2	0.692	−0.013	0.350	0.821	0.393	0.459	0.505	0.453
ELM (Tikhonov)	27000	500	100	25	1651.2	0.692	−0.012	0.349	0.821	0.393	0.459	0.506	0.453
ELM (Box-Cox)	27000	500	100	(25,0.75)	1660.4	0.695	−0.053	0.343	0.819	0.386	0.462	0.512	0.453
ELM (MAE)	27000	500	100	(25,100)	1663.0	0.696	−0.024	0.340	0.818	0.382	0.463	0.516	0.454
ELM (Log-MSE)	27000	500	100	(25,100)	1855.2	0.777	−0.141	0.369	0.774	0.414	0.516	0.578	0.503
ELM (Huber)	27000	500	100	25	1668.9	0.699	−0.024	0.337	0.817	0.379	0.464	0.521	0.455
ELM (L3)	27000	500	100	(10,500)	1773.3	0.743	+0.006	0.413	0.793	0.465	0.493	0.514	0.491
ELM (Corr)	27000	500	100	(25,0.5)	1645.2	0.689	−0.008	0.346	0.822	0.389	0.458	0.505	0.451
ELM (Elastic Net)	27000	500	100	(1,25)	1651.2	0.692	−0.013	0.350	0.821	0.393	0.459	0.505	0.453
ELM (M-Est.)	27000	500	100	25	1700.2	0.712	−0.026	0.337	0.810	0.379	0.473	0.535	0.462
ELM (Lp)	27000	500	100	(25,1.25)	1656.9	0.694	−0.022	0.340	0.820	0.383	0.461	0.513	0.452
ELM (GLM)	27000	500	100	(25,0.1)	1724.8	0.722	+0.011	0.359	0.805	0.404	0.480	0.537	0.474
ELM (Box-Cox roll. 1m)	27000	–	–	(50,0.25)	1786.6	0.747	−0.109	0.363	0.791	0.408	0.497	0.553	0.486
ELM (Box-Cox roll. 6m)	27000	–	–	(50,0.75)	1695.1	0.708	−0.020	0.367	0.812	0.413	0.471	0.519	0.468
ELM (Box-Cox roll. 1y)	27000	–	–	(50,0.75)	1664.6	0.696	−0.044	0.358	0.818	0.402	0.463	0.511	0.459
ELM (Ridge roll. 1m)	27000	–	–	25	1720.3	0.719	+0.004	0.365	0.806	0.452	0.493	0.533	0.493
ELM (Ridge roll. 6m)	27000	–	–	50	1689.6	0.706	+0.005	0.363	0.813	0.449	0.484	0.520	0.485
ELM (Ridge roll. 1y)	27000	–	–	50	1653.7	0.691	−0.016	0.352	0.821	0.436	0.474	0.511	0.474
ELM _{no cyc} (Ridge)	25000	500	100	25	1705.2	0.714	+0.012	0.388	0.809	0.436	0.474	0.521	0.477
NeuralProphet	–	–	–	–	1775.0	0.741	−0.081	0.467	0.794	0.526	0.493	0.526	0.515
NeuralProphet _{no seas.}	–	–	–	–	2054.6	0.858	−0.006	0.553	0.723	0.622	0.571	0.607	0.600

Table 6: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 3$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	4919.5	1.034	−0.082	0.808	−0.270	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.4	0.682	~ 0	0.429	0.448	0.531	0.660	0.738	0.643
BLEND _{corr}	0	–	–	–	2965.9	0.623	−0.029	0.446	0.538	0.552	0.603	0.654	0.603
BLEND _{optim}	48	–	–	–	2701.8	0.568	−0.073	0.382	0.617	0.473	0.549	0.606	0.543
AR-OLS	53	–	–	–	2520.8	0.530	−0.001	0.395	0.667	0.489	0.512	0.548	0.516
ELM (OLS)	27000	500	100	–	2333.7	0.491	−0.011	0.349	0.714	0.432	0.474	0.514	0.474
ELM (Ridge)	27000	500	100	25	2330.8	0.490	−0.013	0.350	0.715	0.433	0.474	0.513	0.473
ELM (Rob. Risk)	27000	500	100	5	2330.8	0.490	−0.013	0.350	0.715	0.433	0.474	0.513	0.473
ELM (Tikhonov)	27000	500	100	25	2330.8	0.490	−0.012	0.349	0.715	0.432	0.474	0.513	0.473
ELM (Box-Cox)	27000	500	100	(25,0.75)	2343.8	0.493	−0.053	0.343	0.712	0.424	0.476	0.520	0.474
ELM (MAE)	27000	500	100	(25,100)	2347.3	0.493	−0.024	0.340	0.711	0.421	0.477	0.524	0.474
ELM (Log-MSE)	27000	500	100	(25,100)	2618.7	0.550	−0.141	0.369	0.640	0.456	0.532	0.586	0.525
ELM (Huber)	27000	500	100	25	2355.7	0.495	−0.024	0.337	0.709	0.417	0.479	0.529	0.475
ELM (L3)	27000	500	100	(10,500)	2503.1	0.526	+0.006	0.413	0.671	0.512	0.509	0.522	0.514
ELM (Corr)	27000	500	100	(25,0.5)	2322.3	0.488	−0.008	0.346	0.717	0.428	0.472	0.513	0.471
ELM (Elastic Net)	27000	500	100	(1,25)	2330.8	0.490	−0.013	0.350	0.715	0.433	0.474	0.513	0.473
ELM (M-Est.)	27000	500	100	25	2399.9	0.504	−0.026	0.337	0.698	0.417	0.488	0.543	0.483
ELM (Lp)	27000	500	100	(25,1.25)	2338.7	0.492	−0.022	0.340	0.713	0.421	0.475	0.521	0.473
ELM (GLM)	27000	500	100	(25,0.1)	2434.6	0.512	+0.011	0.359	0.689	0.445	0.495	0.545	0.495
ELM (Box-Cox roll. 1m)	27000	–	–	(50,0.25)	2520.4	0.529	−0.111	0.361	0.667	0.447	0.512	0.562	0.507
ELM (Box-Cox roll. 6m)	27000	–	–	(50,0.75)	2385.3	0.501	−0.034	0.353	0.702	0.438	0.485	0.526	0.483
ELM (Box-Cox roll. 1y)	27000	–	–	(50,0.75)	2344.4	0.492	−0.057	0.345	0.712	0.427	0.476	0.519	0.474
ELM (Ridge roll. 1m)	27000	–	–	25	2427.1	0.510	+0.004	0.365	0.691	0.452	0.493	0.533	0.493
ELM (Ridge roll. 6m)	27000	–	–	50	2383.7	0.500	+0.005	0.362	0.702	0.449	0.484	0.520	0.485
ELM (Ridge roll. 1y)	27000	–	–	50	2333.1	0.490	−0.016	0.352	0.715	0.436	0.474	0.511	0.474
ELM _{no cyc} (Ridge)	25000	500	100	25	2390.1	0.502	−0.019	0.356	0.700	0.441	0.486	0.529	0.485
NeuralProphet	–	–	–	–	2441.6	0.512	−0.023	0.377	0.688	0.467	0.496	0.532	0.498
NeuralProphet _{no seas.}	–	–	–	–	2783.2	0.584	+0.018	0.425	0.594	0.526	0.565	0.611	0.567

FH = 6 h

Table 7: All data – $FH = 6$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	4786.7	2.004	−0.408	1.103	−0.505	0.730	0.869	0.917	0.839
Persistence P ^o	0	–	–	–	2298.7	0.963	~ 0	0.429	0.653	0.284	0.417	0.494	0.398
BLEND _{corr}	0	–	–	–	2548.0	1.067	−0.181	0.564	0.574	0.373	0.462	0.511	0.449
BLEND _{optim}	48	–	–	–	2164.6	0.906	−0.115	0.423	0.692	0.280	0.393	0.463	0.379
AR-OLS	53	–	–	–	1948.7	0.816	−0.015	0.427	0.751	0.283	0.354	0.402	0.346
ELM (OLS)	27000	500	100	–	1869.1	0.783	−0.020	0.399	0.771	0.264	0.339	0.386	0.330
ELM (Ridge)	27000	500	100	25	1853.8	0.776	−0.024	0.396	0.774	0.262	0.336	0.383	0.327
ELM (Rob. Risk)	27000	500	100	10	1860.2	0.779	−0.024	0.400	0.773	0.265	0.338	0.383	0.329
ELM (Tikhonov)	27000	500	100	25	1854.9	0.777	−0.023	0.395	0.774	0.262	0.337	0.383	0.327
ELM (Box-Cox)	27000	500	100	(25,0.75)	1863.5	0.780	−0.069	0.391	0.772	0.259	0.338	0.386	0.328
ELM (MAE)	27000	500	100	(25,100)	1877.8	0.786	−0.029	0.387	0.768	0.256	0.341	0.394	0.330
ELM (Log-MSE)	27000	500	100	(25,100)	2069.2	0.866	−0.189	0.420	0.719	0.278	0.376	0.431	0.362
ELM (Huber)	27000	500	100	25	1889.5	0.791	−0.031	0.384	0.765	0.254	0.343	0.398	0.332
ELM (L3)	27000	500	100	(10,500)	1975.9	0.827	−0.005	0.464	0.744	0.307	0.359	0.386	0.350
ELM (Corr)	27000	500	100	(25,0.5)	1863.2	0.780	−0.020	0.396	0.772	0.263	0.338	0.385	0.329
ELM (Elastic Net)	27000	500	100	(10,25)	1853.8	0.776	−0.024	0.396	0.774	0.262	0.336	0.383	0.327
ELM (M-Est.)	27000	500	100	25	1912.1	0.801	−0.034	0.382	0.760	0.253	0.347	0.405	0.335
ELM (Lp)	27000	500	100	(25,1.25)	1872.7	0.784	−0.027	0.388	0.770	0.257	0.340	0.392	0.330
ELM (GLM)	27000	500	100	(25,0.1)	1952.4	0.818	+0.012	0.416	0.750	0.275	0.354	0.406	0.345
ELM (Box-Cox roll. 1m)	27000	–	–	(25,0.75)	1941.6	0.811	−0.024	0.421	0.753	0.279	0.352	0.403	0.345
ELM (Box-Cox roll. 6m)	27000	–	–	(100,1)	1902.7	0.795	+0.034	0.441	0.763	0.292	0.345	0.388	0.342
ELM (Box-Cox roll. 1y)	27000	–	–	(50,0.75)	1880.7	0.786	−0.060	0.407	0.768	0.270	0.341	0.388	0.333
ELM (Ridge roll. 1m)	27000	–	–	25	1932.9	0.808	+0.003	0.412	0.755	0.374	0.403	0.436	0.404
ELM (Ridge roll. 6m)	27000	–	–	100	1889.6	0.790	+0.004	0.411	0.766	0.372	0.394	0.422	0.396
ELM (Ridge roll. 1y)	27000	–	–	50	1863.9	0.779	−0.025	0.401	0.772	0.363	0.389	0.418	0.390
ELM _{no cyc} (Ridge)	25000	500	100	25	1909.3	0.799	−0.002	0.439	0.761	0.291	0.347	0.393	0.343
ELM _{no cyc} (Rob.R.)	200	4	2	5	2769.6	0.914	+0.357	0.728	0.534	0.501	0.457	0.450	0.469
NeuralProphet	–	–	–	–	1959.6	0.818	−0.114	0.505	0.748	0.334	0.355	0.399	0.363
NeuralProphet _{no seas.}	–	–	–	–	2190.6	0.915	+0.003	0.554	0.686	0.367	0.397	0.451	0.405

Table 8: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 6$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6756.5	1.420	−0.408	1.103	−1.395	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.389	0.480	0.539	0.469
BLEND _{corr}	0	–	–	–	3596.5	0.756	−0.181	0.564	0.321	0.511	0.532	0.557	0.534
BLEND _{optim}	48	–	–	–	3055.4	0.642	−0.115	0.423	0.510	0.383	0.452	0.505	0.447
AR-OLS	53	–	–	–	2750.6	0.578	−0.014	0.427	0.603	0.387	0.407	0.438	0.411
ELM (OLS)	27000	500	100	–	2638.3	0.555	−0.020	0.399	0.635	0.362	0.390	0.421	0.391
ELM (Ridge)	27000	500	100	25	2616.7	0.550	−0.023	0.396	0.641	0.359	0.387	0.418	0.388
ELM (Rob. Risk)	27000	500	100	10	2625.7	0.552	−0.024	0.400	0.638	0.363	0.389	0.418	0.390
ELM (Tikhonov)	27000	500	100	25	2618.2	0.550	−0.022	0.395	0.640	0.359	0.388	0.418	0.388
ELM (Box-Cox)	27000	500	100	(25,0.75)	2630.4	0.553	−0.069	0.391	0.637	0.355	0.389	0.421	0.388
ELM (MAE)	27000	500	100	(25,100)	2650.5	0.557	−0.029	0.387	0.631	0.351	0.392	0.429	0.391
ELM (Log-MSE)	27000	500	100	(25,100)	2920.8	0.614	−0.189	0.420	0.552	0.381	0.432	0.470	0.428
ELM (Huber)	27000	500	100	25	2667.0	0.561	−0.031	0.384	0.627	0.348	0.395	0.434	0.392
ELM (L3)	27000	500	100	(10,500)	2789.0	0.586	−0.005	0.464	0.592	0.421	0.413	0.420	0.418
ELM (Corr)	27000	500	100	(25,0.5)	2629.9	0.553	−0.020	0.396	0.637	0.359	0.389	0.420	0.390
ELM (Elastic Net)	27000	500	100	(10,25)	2616.7	0.550	−0.023	0.396	0.641	0.359	0.387	0.418	0.388
ELM (M-Est.)	27000	500	100	25	2699.0	0.567	−0.034	0.382	0.618	0.347	0.399	0.442	0.396
ELM (Lp)	27000	500	100	(25,1.25)	2643.4	0.556	−0.027	0.388	0.633	0.352	0.391	0.427	0.390
ELM (GLM)	27000	500	100	(25,0.1)	2755.8	0.579	+0.012	0.416	0.602	0.377	0.408	0.443	0.409
ELM (Box-Cox roll. 1m)	27000	–	–	(25,0.75)	2734.2	0.574	−0.039	0.406	0.608	0.368	0.404	0.439	0.404
ELM (Box-Cox roll. 6m)	27000	–	–	(100,1)	2665.9	0.560	+0.004	0.411	0.627	0.372	0.394	0.422	0.396
ELM (Box-Cox roll. 1y)	27000	–	–	(50,0.75)	2650.3	0.556	−0.072	0.395	0.632	0.358	0.392	0.423	0.391
ELM (Ridge roll. 1m)	27000	–	–	25	2727.0	0.573	+0.003	0.412	0.610	0.374	0.403	0.436	0.404
ELM (Ridge roll. 6m)	27000	–	–	100	2665.9	0.560	+0.004	0.411	0.627	0.372	0.394	0.422	0.396
ELM (Ridge roll. 1y)	27000	–	–	50	2629.7	0.552	−0.025	0.400	0.637	0.363	0.389	0.418	0.390
ELM _{no cyc} (Ridge)	25000	500	100	25	2679.8	0.563	−0.035	0.406	0.623	0.368	0.397	0.429	0.398
ELM _{no cyc} (Rob.R.)	200	4	2	5	3215.1	0.640	+0.171	0.542	0.401	0.471	0.456	0.456	0.461
NeuralProphet	–	–	–	–	2702.9	0.567	−0.043	0.404	0.617	0.367	0.400	0.434	0.400
NeuralProphet _{no seas.}	–	–	–	–	3000.2	0.630	+0.004	0.437	0.528	0.396	0.444	0.489	0.443

FH = 10 h

Table 9: All data – $FH = 10$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	4739.4	1.984	–0.852	1.074	–0.475	0.557	0.740	0.815	0.704
Persistence P ^o	0	–	–	–	2298.9	0.962	~ 0	0.429	0.653	0.223	0.359	0.440	0.341
BLEND _{corr}	0	–	–	–	2775.7	1.162	–0.415	0.592	0.494	0.307	0.433	0.500	0.414
BLEND _{optim}	48	–	–	–	2224.0	0.931	–0.135	0.433	0.675	0.225	0.347	0.423	0.332
AR-OLS	53	–	–	–	1954.4	0.818	–0.016	0.429	0.749	0.222	0.305	0.359	0.295
ELM (OLS)	27000	500	100	–	1886.9	0.790	–0.021	0.406	0.766	0.211	0.295	0.346	0.284
ELM (Ridge)	27000	500	100	25	1879.9	0.787	–0.023	0.405	0.768	0.210	0.294	0.344	0.283
ELM (Rob. Risk)	27000	500	100	10	1878.1	0.786	–0.027	0.407	0.768	0.211	0.293	0.344	0.283
ELM (Tikhonov)	27000	500	100	25	1882.3	0.788	–0.022	0.405	0.767	0.210	0.294	0.345	0.283
ELM (Box-Cox)	27000	500	100	(25,0.75)	1895.9	0.794	–0.070	0.401	0.764	0.208	0.296	0.349	0.284
ELM (MAE)	27000	500	100	(25,10)	1890.9	0.792	–0.030	0.396	0.765	0.206	0.295	0.351	0.284
ELM (Log-MSE)	27000	500	100	(25,100)	2087.1	0.874	–0.192	0.430	0.714	0.223	0.326	0.383	0.311
ELM (Huber)	27000	500	100	25	1910.2	0.800	–0.029	0.392	0.760	0.204	0.298	0.357	0.286
ELM (L3)	27000	500	100	(10,500)	2019.0	0.845	–0.014	0.475	0.732	0.246	0.315	0.350	0.304
ELM (Corr)	27000	500	100	(25,0.5)	1890.9	0.792	–0.019	0.407	0.765	0.211	0.295	0.346	0.284
ELM (Elastic Net)	27000	500	100	(10,25)	1879.9	0.787	–0.023	0.405	0.768	0.210	0.294	0.344	0.283
ELM (M-Est.)	27000	500	100	25	1922.1	0.805	–0.032	0.389	0.757	0.202	0.300	0.361	0.288
ELM (Lp)	27000	500	100	(25,1.25)	1892.1	0.792	–0.028	0.396	0.765	0.205	0.295	0.351	0.284
ELM (GLM)	27000	500	100	(25,0.1)	1967.5	0.824	+0.011	0.423	0.746	0.220	0.307	0.363	0.297
ELM (Box-Cox roll. 1m)	27000	–	–	(25,0.75)	1974.3	0.825	–0.028	0.432	0.744	0.224	0.308	0.363	0.298
ELM (Box-Cox roll. 6m)	27000	–	–	(100,1)	1917.4	0.801	+0.031	0.446	0.759	0.231	0.299	0.347	0.293
ELM (Box-Cox roll. 1y)	27000	–	–	(100,0.75)	1907.4	0.797	–0.066	0.416	0.761	0.216	0.298	0.349	0.287
ELM (Ridge roll. 1m)	27000	–	–	25	1960.2	0.819	–0.003	0.421	0.748	0.391	0.413	0.442	0.415
ELM (Ridge roll. 6m)	27000	–	–	100	1904.9	0.796	~ 0	0.415	0.762	0.386	0.401	0.426	0.405
ELM (Ridge roll. 1y)	27000	–	–	100	1888.1	0.789	–0.025	0.410	0.766	0.382	0.398	0.422	0.401
ELM _{no cyc} (Ridge)	25000	500	100	25	1940.2	0.812	–0.003	0.444	0.753	0.231	0.303	0.354	0.296
NeuralProphet	–	–	–	–	1974.8	0.825	–0.132	0.513	0.745	0.266	0.308	0.357	0.311
NeuralProphet _{no seas.}	–	–	–	–	2175.0	0.908	–0.004	0.554	0.690	0.288	0.339	0.398	0.342

Table 10: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 10$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6688.9	1.406	–0.852	1.074	–1.348	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.399	0.485	0.540	0.475
BLEND _{corr}	0	–	–	–	3917.5	0.823	–0.415	0.592	0.195	0.551	0.586	0.614	0.583
BLEND _{optim}	48	–	–	–	3138.9	0.660	–0.135	0.433	0.483	0.403	0.469	0.519	0.464
AR-OLS	53	–	–	–	2758.3	0.580	–0.016	0.429	0.601	0.399	0.412	0.440	0.417
ELM (OLS)	27000	500	100	–	2663.0	0.560	–0.020	0.406	0.628	0.378	0.398	0.424	0.400
ELM (Ridge)	27000	500	100	25	2653.2	0.558	–0.023	0.405	0.631	0.377	0.397	0.423	0.399
ELM (Rob. Risk)	27000	500	100	10	2650.7	0.557	–0.027	0.407	0.631	0.379	0.396	0.422	0.399
ELM (Tikhonov)	27000	500	100	25	2656.6	0.558	–0.022	0.405	0.630	0.377	0.397	0.423	0.399
ELM (Box-Cox)	27000	500	100	(25,0.75)	2675.7	0.562	–0.070	0.400	0.624	0.373	0.400	0.428	0.400
ELM (MAE)	27000	500	100	(25,10)	2668.7	0.561	–0.030	0.396	0.626	0.369	0.399	0.430	0.399
ELM (Log-MSE)	27000	500	100	(25,100)	2945.6	0.619	–0.192	0.430	0.545	0.400	0.440	0.471	0.437
ELM (Huber)	27000	500	100	25	2696.0	0.567	–0.029	0.392	0.619	0.365	0.403	0.438	0.402
ELM (L3)	27000	500	100	(10,500)	2849.5	0.599	–0.014	0.475	0.574	0.442	0.426	0.429	0.432
ELM (Corr)	27000	500	100	(25,0.5)	2668.8	0.561	–0.019	0.407	0.626	0.379	0.399	0.425	0.401
ELM (Elastic Net)	27000	500	100	(10,25)	2653.2	0.558	–0.023	0.405	0.631	0.377	0.397	0.423	0.399
ELM (M-Est.)	27000	500	100	25	2712.7	0.570	–0.031	0.389	0.614	0.362	0.406	0.443	0.403
ELM (Lp)	27000	500	100	(25,1.25)	2670.4	0.561	–0.028	0.396	0.626	0.368	0.399	0.431	0.400
ELM (GLM)	27000	500	100	(25,0.1)	2776.8	0.584	+0.011	0.423	0.595	0.394	0.415	0.446	0.418
ELM (Box-Cox roll. 1m)	27000	–	–	(25,0.75)	2779.9	0.584	–0.044	0.416	0.595	0.387	0.415	0.446	0.416
ELM (Box-Cox roll. 6m)	27000	–	–	(100,1)	2687.5	0.564	~ 0	0.415	0.621	0.386	0.401	0.426	0.405
ELM (Box-Cox roll. 1y)	27000	–	–	(100,0.75)	2688.6	0.564	–0.076	0.405	0.621	0.377	0.402	0.428	0.402
ELM (Ridge roll. 1m)	27000	–	–	25	2765.6	0.581	–0.003	0.420	0.599	0.391	0.413	0.442	0.415
ELM (Ridge roll. 6m)	27000	–	–	100	2687.5	0.564	~ 0	0.415	0.621	0.386	0.401	0.426	0.405
ELM (Ridge roll. 1y)	27000	–	–	100	2663.8	0.559	–0.025	0.410	0.628	0.382	0.398	0.422	0.401
ELM _{no cyc} (Ridge)	25000	500	100	25	2726.5	0.573	–0.032	0.416	0.610	0.387	0.408	0.435	0.410
NeuralProphet	–	–	–	–	2719.8	0.571	–0.052	0.407	0.612	0.379	0.406	0.437	0.407
NeuralProphet _{no seas.}	–	–	–	–	2986.5	0.627	+0.007	0.441	0.533	0.410	0.446	0.487	0.448